Counterfactual Learning of Stochastic Policies with Continuous Actions

MIPT-UGA AI Workshop

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Logged bandit feedback problem

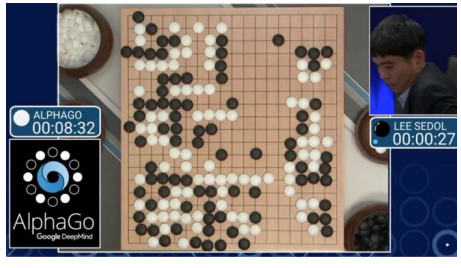
Problem: In interactive systems, full information is not always available.





We wish to learn to **take actions** so as to **maximize a reward** signal...





... while **ensuring safe/affordable learning** for real-world applications with **continuous** treatment (drug dosage, online bidding, etc.).

What are offline contextual bandits?

Multi-armed bandits: For a set of *actions* $a \in [1, K]$, we learn a *policy* π that takes actions $a \sim \pi$ to minimize the *cost* $y(a)$.

Contextual bandits: We add a *contextual* information $x \in \mathcal{X}$ and learn a contextual policy $\pi(\cdot|x)$ to minimize the cost $y(x, a)$.

Online/Offline learning: In an online learning setup, the policy is deployed and the cost signal is observed directly. In the offline setting, the cost is estimated for a given policy.

Counterfactual question: *"What would have been the performance of the system, should the policy had been slightly different?"*

Counterfactual Risk Minimization[1]

Context $x \sim \mathcal{D}_X$ in $\mathcal{X}$, actions $a \sim \pi(\cdot|x)$ and costs $y \sim D_Y(\cdot|x, a)$

Definition

For any policy $\pi \in \Pi$, and feedback cost y, the expected *risk* is:

$$R(\pi) = \mathbb{E}_{(x,a,y)\sim\mathcal{D}_\pi}[y] = \mathbb{E}_{x\sim \Pr(\mathcal{X})}\, \mathbb{E}_{a\sim\pi_0(a|x)}\left[y\frac{\pi(a|x)}{\pi_0(a|x)}\right]$$

which motivates the ***inverse propensity scoring*** (IPS, Horvitz and Thompson, 1952) estimators:

$$\hat{R}_{\text{IPS}}(\pi) = \frac{1}{n}\sum_{i=1}^{n} y_i \frac{\pi(a_i|x_i)}{\pi_{0,i}}.$$

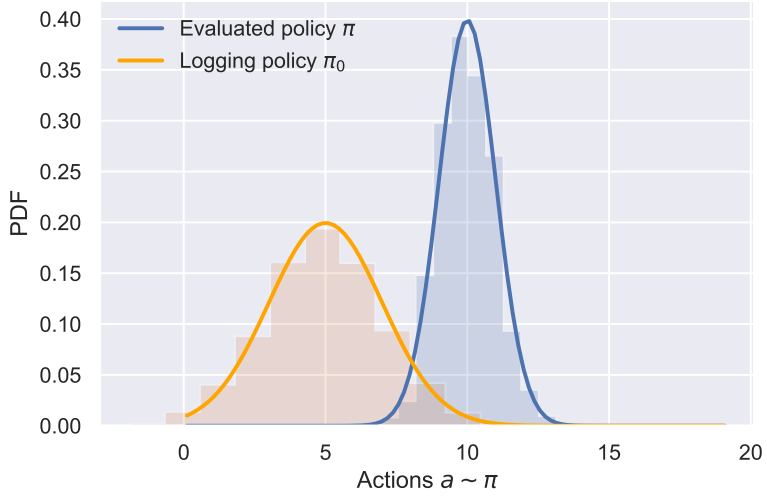
Goal: find a policy π that minimizes $\hat{R}(\pi)$ with regularizers $\Omega(\pi)$.

$$\hat{\pi} \in \arg\min_{\pi\in\Pi} \hat{R}(\pi) + \Omega(\pi).$$

[1] Swaminathan and Joachims, 2015a.

The variance problem

Unbounded variance Evaluated policies π which are "far" from the logging policy and where $\pi_0 \approx 0$ have high variance estimates.



Loss overfitting Unlikely (outlier) samples with $\pi_0 \approx 0$ and high losses can also increase variance.

Solutions Clipped estimators, variance regularizations in CRM learning objectives.

The equivariance problem

The true risk verifies:

$$c + \min_{\pi} \mathbb{E}_{(x,a,y)\sim\mathcal{D}_\pi}[y] = \min_{\pi} \mathbb{E}_{(x,a,y)\sim\mathcal{D}_\pi}[y + c],$$

while the empirical IPS estimator does not:

$$c + \min_{\pi} \frac{1}{n}\sum_{i=1}^{n} y_i w_i^\pi \neq \min_{\pi} \frac{1}{n}\sum_{i=1}^{n} (y_i + c) w_i^\pi$$

where $w_i^\pi = \dfrac{\pi(a_i|x_i)}{\pi_{0,i}}$. When the solution of the optimisation problem is affected by $\delta \leftarrow \delta + c$, the estimator is not equivariant.

Instead,

$$\hat{R}_{\text{SNIPS}}(\pi) = \frac{\sum_{i=1}^{n} y_i w_i^\pi}{\sum_{i=1}^{n} w_i^\pi}$$

is a self-normalized estimator (SNIPS) and verifies it.

Contributions

In this work, we list the following contributions:

- **Modeling:** we propose a new *regularized estimator* with a *contextual modelling* for the *continuous action* case.
- **Optimize:** We show the importance of optimization with *soft-clipping* and *proximal point algorithms*.
- **Evaluate**: We release a new large-scale real-world *dataset*, and show how to perform *offline evaluation*.

Modeling: continuous action setup

In this continous[2] setup, we consider **stochastic** parametrized policies, $\theta = (\theta_\mu, \sigma)$.

Baseline contextual modelling of the mean of the pdf μ (σ std))

- **constant**, context-independent parameterization $\mu = \theta_\mu$
- **stratified** piecewise constant: partition of the feature space $\mathcal{X}$
- **linear** $\mu = \theta_\mu^\top x$, **non-linear** $\mu = \theta_\mu^\top \varphi(x)$

Instead, we could learn a joint representation on (x, a) as loss predictors do in the form of $\eta(x, a) = \theta_\mu^\top \psi(x, a)$

Counterfactual Loss Predictor: we build a stochastic policy $\hat{\pi}_{\text{CLP}}(\cdot|x) = \mathcal{N}(\mu(x), \sigma^2)$ centered around the prediction

$$\mu(x) = \text{argmin}_{a \in \mathcal{A}} \eta(x, a)$$

[2] Kallus and Zhou, 2018.

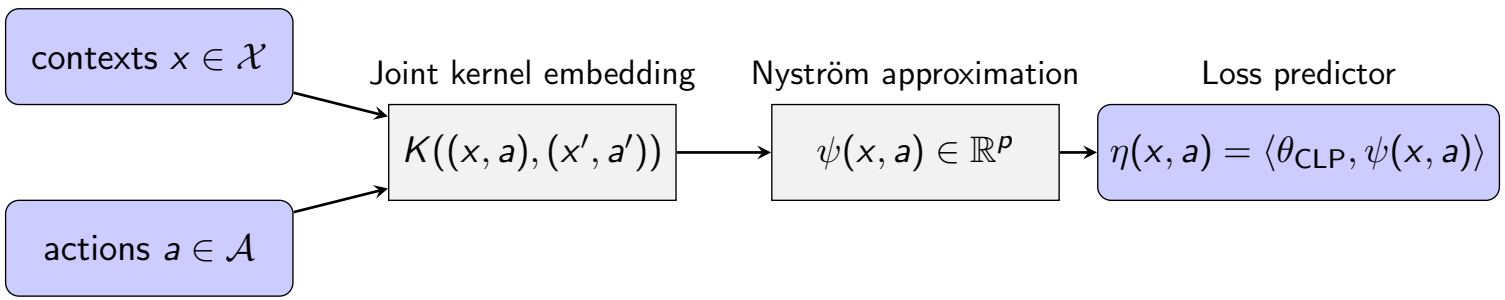
Modeling: the counterfactual loss predictor

Having $\eta(x, a) = \theta_\mu^\top \psi(x, a)$ we use a soft-argmin operator over the action space

$$\mu_{\text{CLP}}(x) = \sum_{i=1}^{m} a_i \frac{\exp(-\gamma \eta(x, a_i))}{\sum_{j=1}^{m} \exp(-\gamma \eta(x, a_j))},$$

by using m anchor points $a_1, \ldots, a_m$ and performing a Nystrom approximation on the kernel action feature map $\psi_{\mathcal{A}}$, where decompose $\psi(x, a) = \psi_{\mathcal{X}}(x) \otimes \psi_{\mathcal{A}}(a)$.

We recapitulate the model

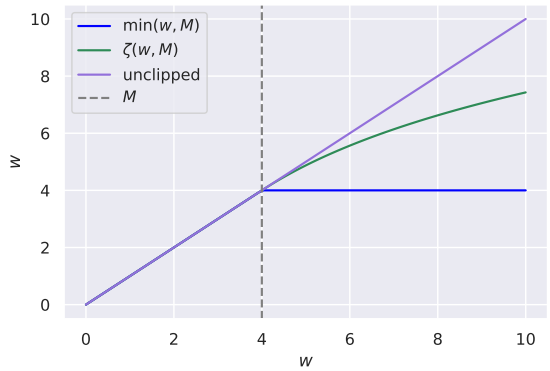


Modeling: a regularization for gradient-based optimization

Counterfactual estimators suffer from **unbounded variance**[3].

Clipping (Bottou et al., 2013) the importance sampling weights:

$$\hat{R}^M_{\text{cIPS}}(\pi) = \frac{1}{n}\sum_{i=1}^{n} y_i \min\left\{\frac{\pi(a_i|x_i)}{\pi_{0,i}}, M\right\}.$$



$$\zeta(w_i, M) = \begin{cases} w_i \\ \alpha(M)\log\left(w_i + \alpha(M) - M\right) \end{cases}$$

where $\alpha(M)\log(\alpha(M)) = M$

Soft-clipping continuous and differentiable operator (scIPS):

$$\hat{R}^M_{\text{scIPS}}(\pi) = \frac{1}{n}\sum_{i=1}^{n} y_i \zeta\left(\frac{\pi(a_i|x_i)}{\pi_{0,i}}, M\right)$$

[3] Bottou et al., 2013.

Optimization: proximal point algorithms

The **learning problem** is non convex (clipping, self-normalized[a], variance penalty, etc.[b]):

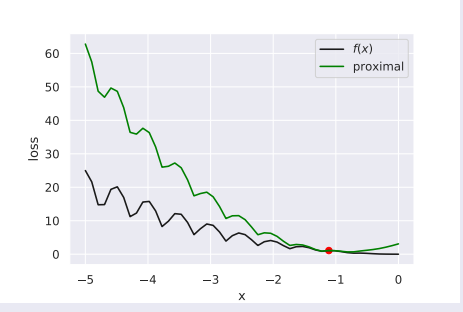
$$\hat{\theta} \in \arg\min_{\theta \in \Theta} \hat{R}(\pi_\theta) + \Omega(\pi_\theta).$$

[a] Swaminathan and Joachims, 2015b.
[b] Dudık, Langford, and Li, 2011.

Proximal Point Method (Rockafellar, 1976)

Approximately solve a sequence of subproblems that are better conditioned and easier to solve than $\mathcal{L}$

$$\theta_k \approx \arg\min_{\theta} \left(\mathcal{L}(\theta) + \frac{\kappa}{2} \|\theta - \theta_{k-1}\|_2^2 \right)$$



Evaluation: releasing a new real-world dataset

Causal advertising setup

For user **features** $x \sim X$ an **action** $a \sim A$ is taken to decide which amount to **bid** B. A **cost** $c \sim C$ and a number of **sales** S define a **reward** R.

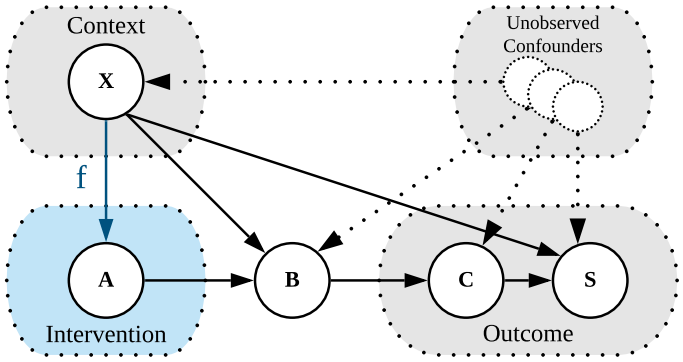


Table: Summary statistics - Criteo Off Policy Continuous Action Dataset

N	d	$\mathbb{E}[R]$	$\mathbb{V}[R]$	$\mathbb{E}[A]$	$\mathbb{V}[A]$
119,971,451	3	11.37	9455.01	1.00	0.01

Evaluation: potential prediction setup

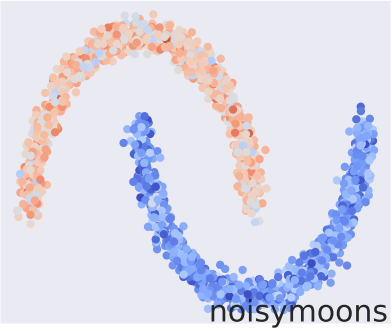


Figure: Example of synthetic 2D features with potentials

Potential prediction setup

Features $x \sim X$ belongs to a **class/group** $g \sim G$ and have **potentials** $p \sim P \mid (G = g)$.

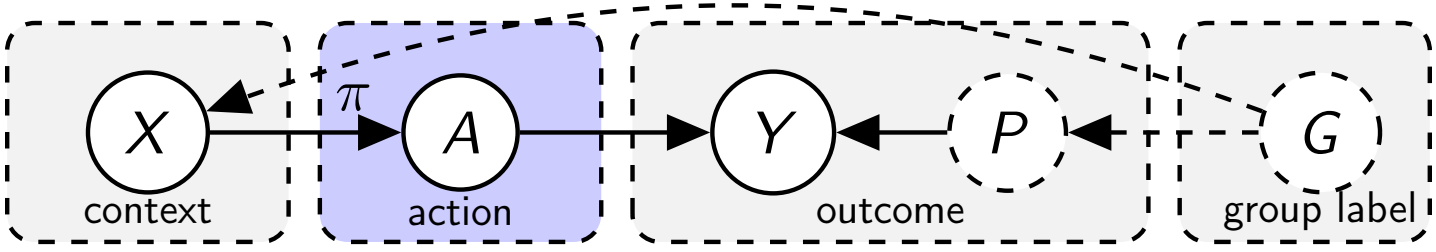


Figure: Causal Graph of the synthetic setting. Depending on the context X, action A should attempt to predict what is the potential P sampled from the unobserved group label G, thus inducing an outcome Y. Unobserved elements are dotted.

Evaluation: semi-synthetic setup and empirical methodology

Warfarin semi-synthetic data: We follow the setup of Kallus and Zhou (2018) on dosage of the Warfarin blood thinner drug. We simulate a contextual bandit environment by using a synthetic reward function that is maximal for actions a that are within 10% of the medical expert's therapeutic drug dosage based on patients covariates.

Methodology We generate training, validation, and test sets and perform model selection with an offline evaluation protocol that uses SNIPS estimates for evaluation and discards policies deemed unsafe with their importance sampling diagnostics and confidence intervals.

Evaluation: discrete vs continuous modelling

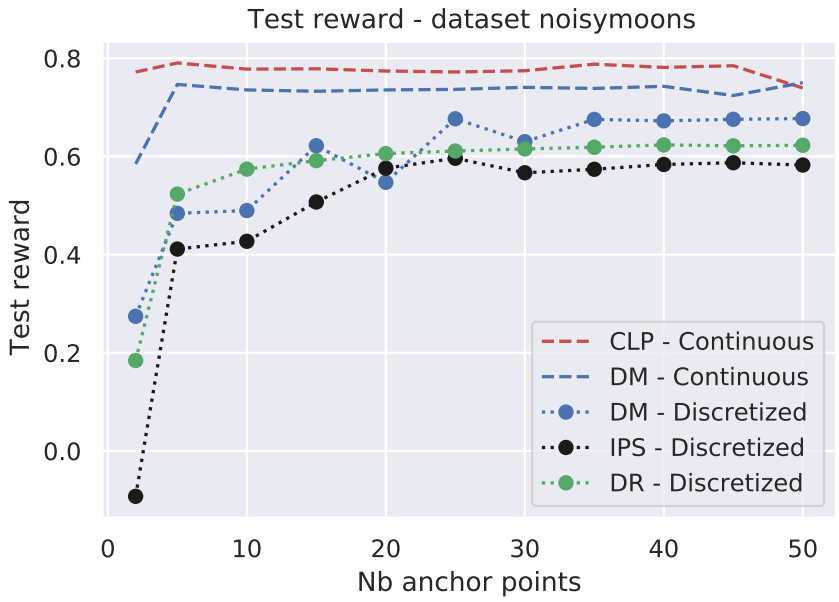


Figure: Continuous vs discretization strategies. Test rewards on NoisyMoons dataset with varying numbers of anchor points for our continuous parametrization for CLP and DM, versus naive discretization.

Naive discretization strategies prevent us from exploiting smoothness assumption on the cost η with respect to actions and from exploiting the continuous structure of the action space.

Evaluation: contextual modelling

Table: Test rewards (higher the better) for several estimators (see main text for details). * refers to the original result from Kallus and Zhou, 2018, denoted by KZ.

	Circles	Moons	GMM	Warfarin	
Constant	0.612	0.612	0.603	−8.964	11.36
Linear	0.612	0.736	0.764	−10.583	10.34
Poly	0.697	0.737	0.745	-	11.13
Chen et al.	0.608	0.645	0.754	-9.407	-
KZ	0.612	0.734	0.785	−10.19*	11.38
CLP	**0.767**	**0.781**	**0.789**	**-8.720**	**11.48**

Overall, we find our CLP parameterization to improve over all baselines, which highlights the effectiveness of the reward predictor at exploiting the continuous action structure.

Evaluation: soft clipping

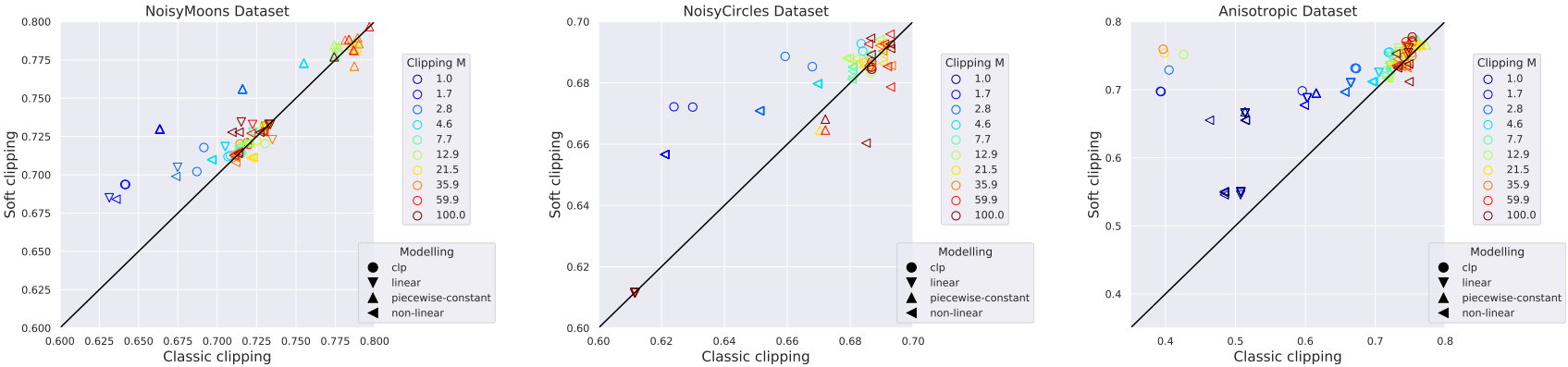


Figure: Influence of soft-clipping. Relative improvements in the test performance for soft- vs hard-clipping on synthetic datasets.

The points correspond to different choices of the clipping parameter M, models and initialization, with the rest of the hyper-parameters optimized on the validation set using the offline evaluation protocol.

Evaluation: proximal point algorithm

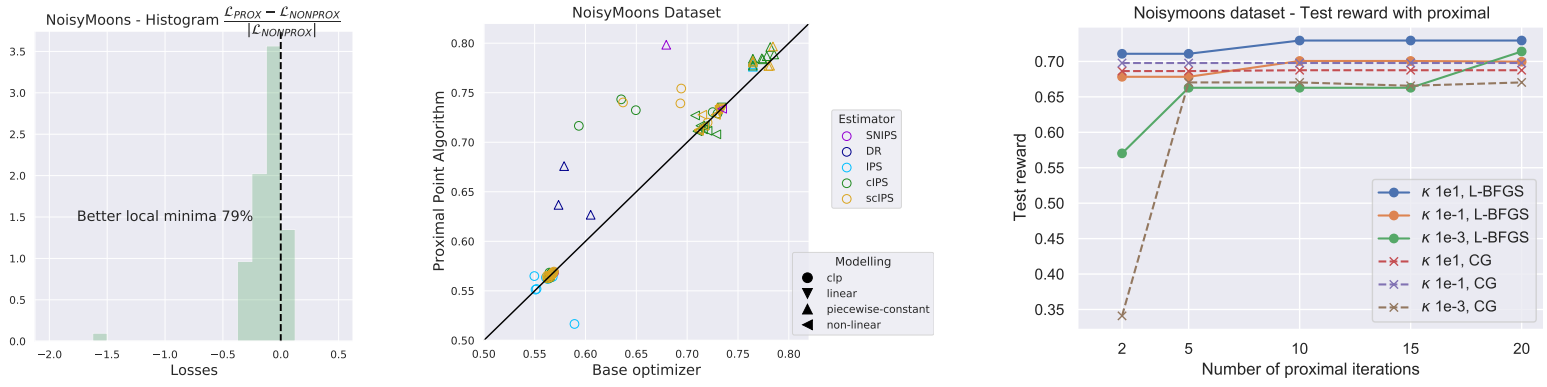


Figure: Influence of proximal point optimization. Relative improvements in the training objective w and w/o using the proximal point method (left), relative improvements in the test performance w and w/o using the proximal point method (center) comparison of test performance with different number of proximal steps and (right).

Take-home message and future work

Take-home message:

- **Modeling** on the continuous action space requires more than discretization strategies. Our proposed CLP modeling achieves better empirical results.
- **Optimization** is equally important as **modeling** for the resulting non-convex cost functions. To that effect, we introduce smooth clipping, and use proximal point algorithms.
- **Evaluation** for real-world offline datasets has received little attention in continuous settings: we provide a real-world dataset and a protocol for performance measurement.

Future research:

- Doubly robust estimator with the CLP model

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